Worksheet 5

Time series – DETAILED SOLUTIONS

Badr ABOUFARASSE, PhD. – Academic year 2026–2027

Detailed solution – Exercise 1

General trend $\to T$; recession movement $\to C$; recurring peak $\to S$; exceptional shock $\to R$.

$$\boxed{Y = T + C + S + R} \quad \text{(additive)},$$

$$\boxed{Y = T \times C \times S \times R} \quad \text{(multiplicative)}.$$

Later in the course, $C = R = 0$ in the additive model and $C = R = 1$ in the multiplicative model.

Detailed solution – Exercise 2

Simple 4-term moving averages are $24.75, 26.50, 29.00, 31.00, 34.25$. Centred values are

$$(24.75 + 26.50)/2 = \boxed{25.625},$$

$$(26.50 + 29.00)/2 = \boxed{27.75}, \quad (29 + 31)/2 = \boxed{30.00},$$

$$(31 + 34.25)/2 = \boxed{32.625}.$$

An even-order simple moving average falls between two periods, so it must be centred.

Detailed solution – Exercise 3

$T_1 = 18.4$ and $T_2 = 21.4$. Therefore

$$S_1 = 18/18.4 = 0.978 \approx 0.98, \quad S_2 = 26/21.4 = 1.215 \approx 1.21.$$

Seasonal ratios are averaged by quarter. The rounded means sum to 4.11, so their mean is 1.0275. Corrected coefficients are

$$\boxed{0.87591,\ 1.11922,\ 0.78832,\ 1.21655}.$$

Forecasts:

t	Trend	Seasonal coefficient	Seasonalised forecast
13	54.4	0.87591	47.65
14	57.4	1.11922	64.24
15	60.4	0.78832	47.61
16	63.4	1.21655	77.13

The deseasonalised forecast is simply the trend value.